AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Microsoft EAT02, WA -- station KEAT, America/Los_Angeles
Committed pair	OSM 1173537116 -> 1173537117 Microsoft EAT03 / Microsoft EAT04
Facade gap	116.3 m between the two halls
Weather record	43,269 real hourly records from KEAT
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.6164 C at 115 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-09-10 (chatter)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 2 mode changes. That is 6 more than the reactive incumbent, at 0 unsafe hours against its 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 8 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	1 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 8 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dry-bulb	21.727	21.0	21.670	0.904
01	mechanical	no	dry-bulb	21.464	21.0	21.110	0.843
02	mechanical	no	dry-bulb	21.436	21.0	21.110	0.766
03	FREE	yes	-	20.614	21.0	19.440	0.749
04	FREE	yes	-	20.677	21.0	20.000	0.780

05	FREE	yes	-	20.133	21.0	18.890	0.708
06	FREE	yes	-	18.986	21.0	17.780	0.671
07	FREE	yes	-	20.969	21.0	19.440	1.033
08	FREE	yes	-	20.621	21.0	17.780	1.366
09	FREE	yes	-	20.894	21.0	18.330	1.343
10	mechanical	no	dry-bulb	21.439	21.0	17.220	1.266
11	mechanical	no	dry-bulb	21.469	21.0	18.330	1.482
12	mechanical	yes	-	20.614	21.0	16.110	1.535
13	mechanical	yes	switch budget	19.877	21.0	15.833	1.608
14	mechanical	yes	switch budget	19.877	21.0	15.833	1.445
15	mechanical	yes	switch budget	17.939	21.0	16.110	1.080
16	mechanical	yes	switch budget	17.306	21.0	16.670	0.973
17	mechanical	yes	switch budget	16.601	21.0	16.127	0.997
18	mechanical	no	dew point	15.629	21.0	15.000	0.921
19	mechanical	no	dew point	16.165	21.0	15.560	1.322
20	mechanical	no	dew point	15.959	21.0	15.560	1.601
21	mechanical	no	dew point	15.328	21.0	15.560	1.440
22	mechanical	yes	-	15.605	21.0	15.560	1.225
23	mechanical	yes	-	15.887	21.0	16.110	1.011

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.727 C against a 21.0 C limit, so it fails by 0.727 C. It would take a limit 0.727 C higher, or a bound 0.727 C tighter, to change this hour.

01:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.464 C against a 21.0 C limit, so it fails by 0.464 C. It would take a limit 0.464 C higher, or a bound 0.464 C tighter, to change this hour.

02:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.436 C against a 21.0 C limit, so it fails by 0.436 C. It would take a limit 0.436 C higher, or a bound 0.436 C tighter, to change this hour.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.614 C, which is 0.386 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.749 C, and the margin is measured, not chosen: 0.689 C of group-conditional forecast error for this hour of day, plus 0.0591 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.677 C, which is 0.323 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.780 C, and the margin is measured, not chosen: 0.658 C of group-conditional forecast error for this hour of day, plus 0.1222 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.133 C, which is 0.867 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.708 C, and the margin is measured, not chosen: 0.575 C of group-conditional forecast error for this hour of day, plus 0.1327 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.986 C, which is 2.014 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.671 C, and the margin is measured, not chosen: 0.575 C of group-conditional forecast error for this hour of day, plus 0.0959 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.969 C, which is 0.031 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.033 C, and the margin is measured, not chosen: 0.895 C of group-conditional forecast error for this hour of day, plus 0.1386 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.621 C, which is 0.379 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.366 C, and the margin is measured, not chosen: 1.300 C of group-conditional forecast error for this hour of day, plus 0.0661 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.894 C, which is 0.106 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.343 C, and the margin is measured, not chosen: 1.284 C of group-conditional forecast error for this hour of day, plus 0.0588 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.439 C against a 21.0 C limit, so it fails by 0.439 C. It would take a limit 0.439 C higher, or a bound 0.439 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.469 C against a 21.0 C limit, so it fails by 0.469 C. It would take a limit 0.469 C higher, or a bound 0.469 C tighter, to change this hour.

12:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.614 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.877 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

14:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.877 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.939 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.306 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.601 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.629 C would have passed. The outside air is too HUMID: the dew-point bound is 15.23 C against a 15.0 C maximum, failing by 0.23 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

19:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.165 C would have passed. The outside air is too HUMID: the dew-point bound is 15.28 C against a 15.0 C maximum, failing by 0.28 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

20:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.959 C would have passed. The outside air is too HUMID: the dew-point bound is 15.55 C against a 15.0 C maximum, failing by 0.55 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

21:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.328 C would have passed. The outside air is too HUMID: the dew-point bound is 15.28 C against a 15.0 C maximum, failing by 0.28 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

22:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.605 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.887 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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